Final Project: UGV

1. Overview

What is UGV?

A **UGV (Unmanned Ground Vehicle)** is a ground robot with **no driver on board**. It senses its surroundings through sensors and decides how to move **by itself**, using a program that you write.

Every group will design, build, and test its own UGV: a ground robot with an onboard camera that can sense and act on its own.

A real-life example: a robot vacuum cleaner(just an example it's not what we are doing). Nobody drives it. Its sensors detect walls and furniture, its small computer decides where to turn, and its motors move it around the room, all on its own battery. Our UGV works the same way, except that instead of cleaning, it uses a camera to find a flag and drives toward it.

How your UGV works

Your UGV has four main abilities:

- **Sense:** the camera (ESP32-CAM) sees the flag, and a distance sensor measures how far away it is.
- **Think:** the ESP32 runs the control program that decides what to do.
- **Act:** the motors and motor driver move the vehicle.
- **Stand alone:** it runs from its own battery, with nobody controlling it.

The big picture

The camera looks for the flag and works out its color. The vehicle announces that color, then drives toward the flag and holds a fixed distance of **50 cm** from it. All four teams must do their part well, because the vehicle only works if the control, electrical, mechanical, and vision parts work together.



2. Group Structure

Each group consists of **four teams**. Every team is responsible for one part of the vehicle.

Team	Responsibility	What this team should produce
Control	Programming, communication, and the vehicle's control logic (including the PID controller that keeps the 50 cm distance)	Commented code, a record of how the controller was tuned, and a record of the code versions
Electrical	Wiring, power, motor drivers, and connecting the electronics	A wiring diagram, a list of which pin connects to what, and a safe, stable power design
Mechanical	The chassis, wheels/tracks, motor mounting, and the overall build	Sketches or CAD, a list of parts, photos of the build, and mounts for the camera and sensor
Computer Vision	Processing the ESP32-CAM video feed (for example detection and tracking)	The detection method, the settings used, and test results under different conditions

Members in more than one subteam: a member who belongs to more than one subteam can also work on the other subteam's tasks, but **only after finishing the tasks of their main subteam**.

3. Project Requirements

Mandatory components

Component	What it does in the project
ESP32	The main controller. It runs the control logic, reads the distance sensor, and drives the motors.
ESP32-CAM	The camera. It captures video so the flag can be detected and tracked.
Ultrasonic distance sensor	Measures the distance to the flag so the vehicle can keep 50 cm.

All groups must use an ultrasonic sensor to measure the distance to the flag.

Programming software: the ESP32 must be programmed using the Arduino IDE.

Weight limit: the complete UGV (chassis, motors, battery, and all electronics) must weigh 2 kg or less. A lighter UGV is considered creative and earns bonus points, as long as the main mission works first.

Justify every component

For **each component** your team uses, you must document **why you needed it**. The justification must answer three questions:

1. **What does it do** in our system?
2. **What alternatives** did we consider?
3. **Why did we choose this one** instead of the alternatives?

A short paragraph or a small comparison table is enough. **A component without a written justification counts as incomplete work.**

4. Mission

The UGV must **detect a colored flag and announce which color it is**. It must then **use a sensor to track the flag and stay 50 cm away from it**.

The mission step by step

1. **Detect:** the UGV finds the flag using the camera.
2. **Announce:** the UGV tells us the flag's color. How it announces the color is up to your team, so be creative: creativity earns bonus points.
3. **Track:** the flag stays fixed in one place, and the UGV drives forward toward it, keeping it centered in the camera view.
4. **Hold 50 cm:** the UGV stops at 50 cm from the flag, measured with the ultrasonic sensor.

The flag: color and size

- The **color and size of the flag will be announced on the day of the competition.**
- Do **not** build your system around one assumed color or size.
- Design your system so it can be **adjusted quickly on the day**. For example, make the color settings easy to change or calibrate without rewriting the whole program, and do not hard-code the flag's size.
- During practice, **test with several different colors and sizes** so you are ready for whatever is announced.

How the competition works

1. **Announcement:** on competition day, the flag's color and size are announced to all groups.
2. **Preparation:** each group gets a short preparation time to adjust and calibrate its vehicle for that flag.
3. **Run:** each group runs its UGV in front of the judges. The UGV detects the flag, announces its color, drives forward toward the fixed flag, and stops at 50 cm from it.
4. **Judging:** the judges assess the result and the documentation.

5. What We Expect from Each Group

- Build the vehicle and assemble the electronics.
- Program the ESP32 and ESP32-CAM so the UGV can detect the flag, announce its color, and keep 50 cm from it.
- Test and improve the vehicle until it works reliably.
- Document every step of the project: design decisions, **the justification for each component**, wiring, code versions, test results, problems you faced, and how you fixed them. Notes, photos, and screenshots are welcome.

How you will be graded

Grading is based on two things:

1. **How well each subteam applied what it learned** in its own area.
2. **The final result** of the vehicle on competition day.

A vehicle that works but is not backed by solid subteam work will **not** receive full marks.

Category	Points
Control	10
Electrical	10
Mechanical	10
Computer Vision	10
Competition day (final result)	10
Total	**50**
Bonus: creative ideas	Bonus points

- **Documentation counts inside each team's grade.** Each team is judged on its own documentation, including its component justifications.
- **Bonus:**

  Creativity is noticed and appreciated, as long as the main mission works first.

6. Frequently Asked Questions

Which flag will be used?
The flag's color and size will be announced on the day of the competition. Prepare for different colors and sizes.

Can we build for one color only?
No. Your system must be adjustable on the day, so a different color or size can be handled quickly.

How do we announce the color?
The method is up to your team. Be creative: creativity earns bonus points, as long as the main mission works first.

Which distance sensor should we use?
Use an ultrasonic sensor. You must still document why you needed it and why it suits this project.

Which software do we use to program the ESP32?
You must use the Arduino IDE to program the ESP32.

How heavy can the UGV be?
The complete UGV must weigh 2 kg or less, including the battery and all electronics. A lighter UGV is considered creative and earns bonus points, as long as the main mission works first.

What does "track" mean?
The flag stays fixed in one place. The UGV keeps the flag centered in the camera view, drives forward toward it, and stops at 50 cm from it.

Do we need to justify every component?
Yes. Every component your team uses needs a written justification (what it does, the alternatives, and why you chose it).

What if our vehicle works perfectly but our documentation is weak?
You will not receive full marks. Grading covers both the subteam work and the final result.

Can I help another subteam?
Yes, but only after finishing the tasks of your own main subteam.

Does creativity matter?
Yes. Creativity earns bonus points.